

Cross-dataset 2 Results and Analysis

We train and test our dataset for:

GRU (Gated Recurrent Unit), Transformer and GAN (Generative Adversarial Network)

Train Dataset: dec2025rb2r2_updated.txt

Test Dataset: feb2026rb2r2_updated.txt

Summary:

For this dataset, both GRU and Transformer perform strongly, with high R^2 values (>0.85) for both voltage and power, while GRU achieves the best overall performance ($R^2 \sim 0.92$). Transformer follows closely but shows slightly higher errors, especially in power prediction. In contrast, GAN performs poorly, with highly negative R^2 for voltage and very low R^2 for power, indicating unreliable predictions. This highlights that **temporal models (GRU) are most effective, followed by attention-based models, while GAN is unsuitable.**

Voltage GRU Results:

MAE: 0.06499135878731901

RMSE: 0.09529210226208853

R2: 0.919829712562731

Power GRU Results:

MAE: 6.273252518318047

RMSE: 39.37541377762526

R2: 0.9219152141936725

Voltage Transformer Results:

MAE: 0.08255038787708871

RMSE: 0.10726554977570141

R2: 0.8984177712225073

Power Transformer Results:

MAE: 23.788013610710365

RMSE: 54.51630603195824

R2: 0.8503208389853638

Voltage GAN Results:

MAE: 2.8532904546803044

RMSE: 3.134009843773177

R2: -85.71573783450533

Power GAN Results:

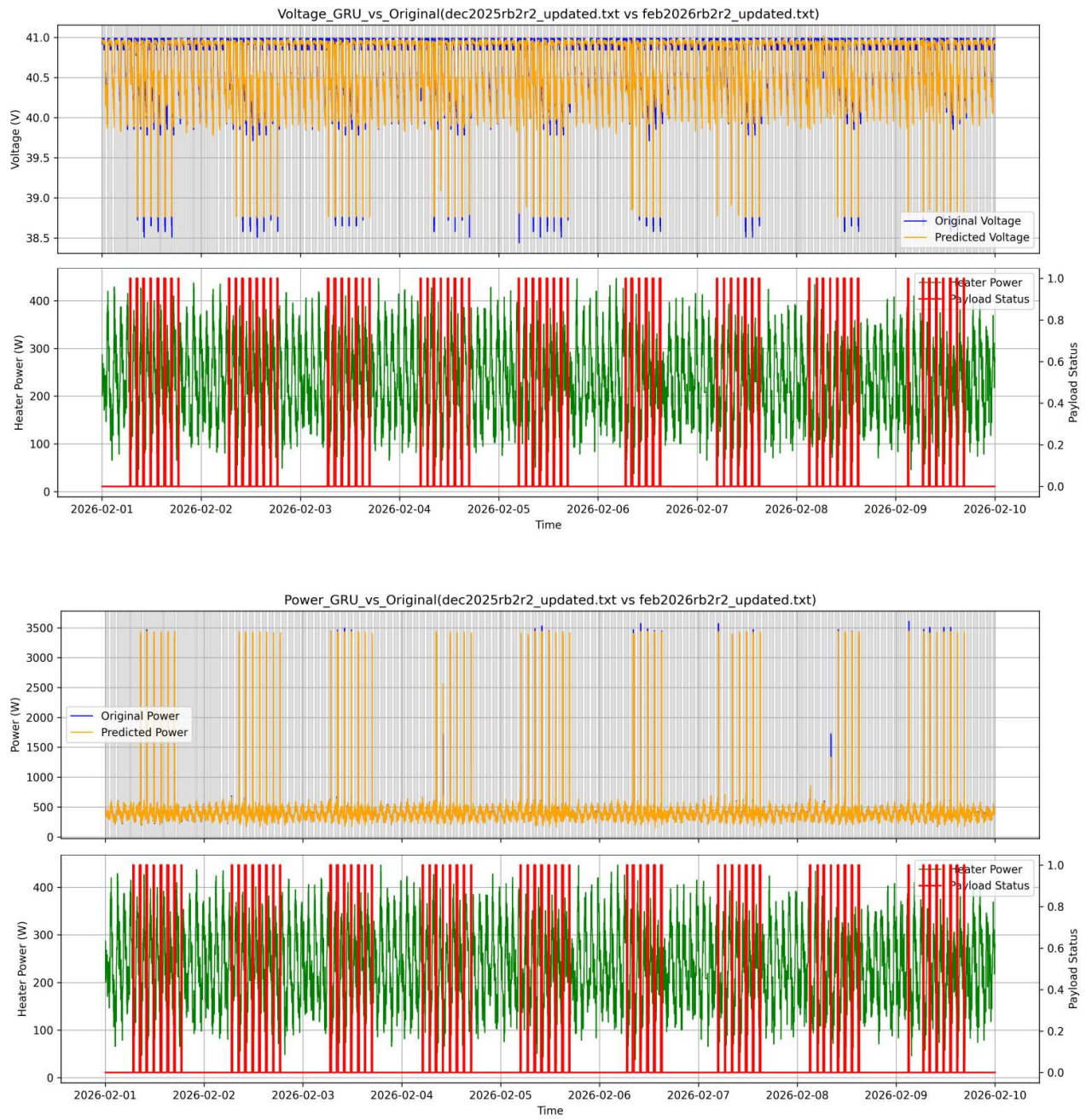
MAE: 64.49836965540396

RMSE: 136.83123425328404

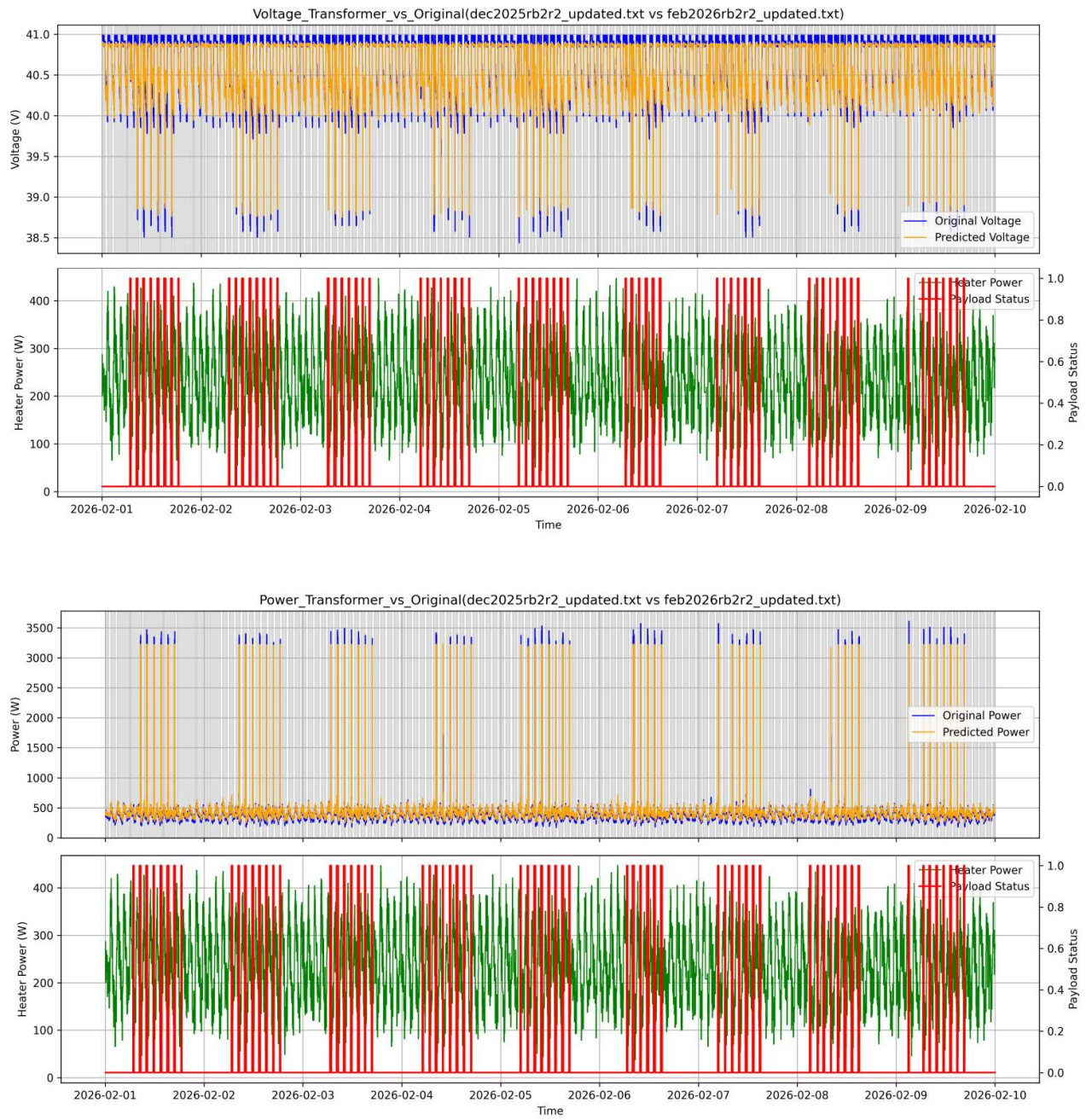
R2: 0.05707101166298634

Plots:

1. GRU



2. Transformer



3. GAN

